UNIVERSITY OF LEEDS

Module Handbook

OCOM5204 Data Mining and Text Analytics (23/24 January F02)

Contents

Module Handbook 1
Contents 1
Accessibility 2
Welcome to the Module 3
Module Team 3
Module Summary 3
Learning Outcomes 3
Syllabus 4
Unit 1: Introduction to corpus linguistics, text analytics and Sketch Engine 4
Unit 2: Text data mining and Weka 4
Unit 3: Word meanings, text tagging, and scaling to large corpora 4
Unit 4: Machine translation, information extraction, and Python tools fortext analytics 4
Unit 5: Chatbots and text analytics for university education 5
Unit 6: Current research in text analytics, and BERT 5
Unit 7 Assessment 5
Key Dates 6
Module Assessment 7

© University of Leeds

Assessment 1: Multiple Choice Questions....8

Assessment 2: Report - Proposal for a research project 11

Reading List 17

Additional Information and Support 17

Key Documents 17

Skills@Library 17

General Module Forum 17

Student Success Advisor 17

Recording Policy 18

Complaints 18

Accessibility

For information in alternative formats please email DESstudents@leeds.ac.uk or speak with your Student Success Advisor.

© University of Leeds

Welcome to the Module

Welcome to OCOM5204M Data Mining and Text Analytics. This handbook contains administrative information related to the module to help you to organise study time for the module. We hope that you find this material useful and the work you do on themodule rewarding.

Module Team

Your module leader is Noorhan Abbas. You will have the opportunity to speak to Noorhan during the live weekly webinar sessions. If you need to contact the module leaderdirectly, you can find their details in the 'Staff Information' section of Minerva.

Module Summary

This module will provide an introduction to linguistic theory and terminology. Over thecourse of this module, you will develop an understanding of, and the ability to use, algorithms and resources for implementing and evaluating text mining and analytics systems. You will learn how to develop solutions using open-source and commercial toolkits, and will explore the applications of data mining and text analytics through case studies in information retrieval and extraction.

Learning Outcomes

Successful completion of this module will help you to:

1. understand theory and terminology of empirical modelling of natural language
2. understand and use algorithms, resources and techniques for implementingand evaluating text mining and analytics systems
3. demonstrate familiarity with some of the main text mining and analyticsapplication areas
4. appreciate why unrestricted natural language processing is still a majorresearch task.

© University of Leeds

Syllabus

Unit 1: Introduction to corpus linguistics, text analytics and Sketch Engine

In this unit, you will learn about corpus linguistics background, terminology and applications. The unit will also introduce you to the online text analytics tool SketchEngine, an important tool for corpus management, including text collection and analysis. You will have the opportunity to practise using Sketch Engine and WebBootCat for text corpus data-set capture and analysis.

Unit 2: Text data mining and Weka

This unit will introduce you to another important data mining tool, Weka. You will learn about N-gram models and classifiers for text data mining, and have the opportunity to practise using Weka for a range of tasks. You will also explore someonline communities for data mining and text analytics professionals.

Unit 3: Word meanings, text tagging, and scaling to large corpora

In this unit, you will be introduced to numerical vector representations of word meanings. You will learn about methods and applications for text tagging such asPart of Speech (PoS) tagging and Named Entity Recognition (NER), and explore research on scaling from small to large data-sets for practical applications of datamining and text analytics.

Unit 4: Machine translation, information extraction, and Python tools fortext analytics

This unit will introduce you to challenges and methods for machine translation and for information extraction. You will also learn about hand-crafted AI rule-based systems as an alternative approach to machine learning from a large data-set. Youwill then examine a range of Python tools for text analytics.

© University of Leeds

Unit 5: Chatbots and text analytics for university education

In this unit, you will be introduced to methods and applications for chatbots and dialogue systems. In particular, you will examine practical uses of chatbots, such ascollecting student feedback at universities. You will then have the opportunity to explore and try a range of different chatbot systems, including the LOLA and Hubertchatbots.

Unit 6: Current research in text analytics, and BERT

In this final unit, you will learn about methods for information retrieval, and challenges and methods for dealing with Multi Word Expressions in text analytics. You will also see how to extend language models based on word embeddings to non-text annotations like emojis. You will be introduced to BERT, a method and toolkit from Google Labs for understanding meaning relationships between sentences. You will then have the opportunity to discuss applications of BERT, andany NLP tasks not suitable for BERT-like deep learning.

Unit 7 Assessment

In this unit of the module you will submit your final assessment based on learningfrom this module. This activity will help consolidate your learning from the moduleand will provide you with a basis for further study, work or practice.

© University of Leeds

Key Dates

	Monday	Tuesday	Wednesday	Thursday	Friday
	8 January Module starts Assessment 2 release	9	10	11	12
Unit 1	15 Unit 1 webinar 14:00 – 15:00 UTC (London) 16:00 – 17:00 UTC (London)	16	17	18	19
Unit 2	22 Unit 2 webinar 14:00 – 15:00 UTC (London) 16:00 – 17:00 UTC (London)	23	24	25	26
Unit 3	29 Unit 3 webinar 14:00 – 15:00 UTC (London) 16:00 – 17:00 UTC (London)	30	31	1 February	2
Unit 4	5 Assessment 1 release Unit 4 webinar 14:00 – 15:00 UTC (London) 16:00 – 17:00 UTC (London)	6	7	8	9
Unit 5	12 Assessment 1 deadline Unit 5 webinar 14:00 – 15:00 UTC (London) 16:00 – 17:00 UTC (London)	13	14	15	16
Unit 6	19 Unit 6 *extended* webinar with Q&A on assessments	20	21	22	23

© University of Leeds

	14:00 – 15:30 UTC (London) **16:00 – 17:30 UTC (London)**				
Study break	**26** **Assessment 2 deadline**	**27**	**28**	**29**	**1 March**

Please see the module area in Minerva for specific timings for webinars and assessment submission deadlines.

Module Assessment

You should familiarise yourself with the Digital Education Service Code of Practice on Assessment (CoPA) before you start your first assignment. The CoPA contains importantinformation about marking, late submissions, resits, mitigating circumstances and more.

© University of Leeds

Assessment 1: Multiple Choice Questions

Summary

This is a multiple choice answer (MCQ) assessment, testing you on your learning from Units 1, 2, 3 and 4.

Deadline

The MCQ test will be accessible for one week, from 08:30am UTC (London) on Monday 5th February.

Due: Monday 12th February 8:30am UTC (London)

Please note: There will be no late submissions for this assignment unless you have been granted an extension.

Purpose

The purpose of this assignment is to help you to evaluate your understanding of the main concepts, theories and terminology covered in Units 1, 2, 3 and 4. It will also assess your ability to apply the methods you have studied in these units.

Key Information

The assessment has a time limit of 60 minutes. This will enable you to practice and apply your learning from the module.
The assessment is comprised of 30 multiple choice questions.

Task

You will have 60 minutes to complete a series of multiple choice questions based on your learning from Units 1, 2, 3 and 4.

The test is comprised of 30 multiple choice questions.

Each question has several suggested answers, and for each possible answer you must select the correct answer.

© University of Leeds

You will need

Before you start the test, please have ready:

- a pen and paper
- a calculator

Academic Integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

The assessment will be completed on Minerva and the link to access the test will be available once the test window is open.

- The test will be available for one week. During that week you will have one attempt at the test.
- You have a **time limit of 60 minutes** to complete the test once you have begun.
- The timer keeps running whether or not you're actively working in the Test. Once you start, if you save a draft or leave the Test page, the countdown continues and your work is saved and submitted automatically when time is up.
- Should you leave the page or the test times out before the deadline, you can reload the module page and return to the test. It will resume from where it was last autosaved.
- You may change any of your answers until your test is submitted.
- Please review your answers carefully before submitting your test as once you have made the submission no amendments will be possible. Your answers will be submitted when you select the submit button at the bottom of the screen.
- Please note that Minerva might display a score once you submit your answers.

© University of Leeds

However, this might be different from the final score you are awarded after marking.

- **Pressing the Enter button on the keyboard or reloading the page after beginning the test will also submit your answers and lock the test so take care not to do so as this may result in you failing to complete the test.**
- Mistakes and problems with submission cannot normally be rectified once you have submitted the test.

Marking

This is a compulsory assessment for OCOM5204M Data Mining and Text Analytics. It is worth 30% of your overall module grade. Each question carries 1 mark, with a total of 30 marks available.

Feedback

In line with normal practice for exams, only the overall mark for the test will be available. Your grade will be visible via 'Gradebook' on Minerva once marking has been completed.

© University of Leeds

Assessment 2: Report - Proposal for a research project

Summary

Develop and write a proposal for a research project using data mining and text analytics theories and techniques for a practical application of your choice.

Deadline

Due: Monday 26th February at 8:30am UTC (London)

Purpose

The purpose of this assignment is for you to investigate how to apply AI methods, resources and techniques for implementing and evaluating data mining and text analytics in a practical applied research project, and then to summarise and present your knowledge in a research proposal report.

Key Information

Develop and write a proposal for a 6-month research project using data mining and text analytics theories and techniques for a practical application of your choice.

Task

You are asked to develop a 6-month research project proposal, using data mining and text analytics theory, methods and technologies for a practical application of your choice. Lectures and online learning resources will include examples of data mining and text analytics methods, techniques, resources, and applications. You can also include other tools and techniques in your maximum 9 pages in total.

When writing an applied data mining and text analytics research proposal, you can learn from advice provided by the Engineering and Physical Sciences Research Council. EPSRC is a major UK supporter and funder for research projects in Engineering and Physical Sciences,

© University of Leeds

including AI. The EPSRC website gives guidance on writing research project proposals, see https://epsrc.ukri.org/funding/applicationprocess/preparing/writing/
https://epsrc.ukri.org/funding/applicationprocess/preparing/writing/caseforsupport/

A standard proposal to EPSRC includes several forms and tables, but the core document is "**Proposed research and its context**" to include:

- Research hypothesis & objectives,
- Background
- Importance and contribution to knowledge
- Pilot study
- Programme and methodology
- And Workplan diagram, eg Gantt Chart

You should use these headings to structure your report, and write a section for each of these:

- 2 pages for Pilot study and Programme and methodology
- 1 page each for other sections (listed above)
- plus 1 page for References

Maximum of 9 pages in total

In addition, as an Appendix of up to 3 pages, describe your use of data mining and text analytics tools in developing your Report. This could include: tools used in the small pilot study to trial the methods proposed; use of tools like Google Scholar or ChatGPT in searching for background information and drafting the report (include examples of query and results).

General Guidance and Study Support

See Minerva Learning Resources in the module for guidance and study support on the sections of the **Proposed Research and Context**:

Research hypothesis and objectives: Set out your research idea or hypothesis. Explain why the proposed project is novel and timely, e.g. emphasising the scientific ambition, or any potential
transformative outcomes. Identify the overall aims of the project and the measurable objectives against which the outputs, outcomes and impacts of the work will be assessed.

Background: Introduce the background to the proposal and explain its context. Explain how

© University of Leeds

this work relates to past research and current projects at Leeds University, in the UK and abroad.

Importance and contribution to knowledge: Explain how the project may contribute to current or future economic success; to future development of key emerging industries; or addresses key societal challenges. Describe how your research would benefit national and international research, and engage with research in other disciplines to broaden the reach of the new knowledge.

Pilot study: Describe your implementation of an initial pilot study to demonstrate feasibility of the project, including selection of a simple case study, acquiring sample data, development and evaluation of your prototype solution.

Programme and methodology: Describe the work programme including research and user evaluation. Identify the contribution of each member of the research team including users and/or stakeholders. Provide milestones and deliverables that you will use to monitor progress, and explain how the project will be managed.

The research work programme should make use of an appropriate methodology for AI projects, such as CRISP-DM; and should include use of at least two data mining and/or text analytics methods, tools or techniques introduced in the module (eg SketchEngine, Weka, ChatGPT)

1-page workplan diagram must match the written description of the research work programme, showing start, end and duration of each phase or work-package.

Presentation

For this report, there are presentation requirements you need to follow:

- There is a strict page limit. You can only submit to 12 pages in total (See 'Task' section for more information on this)
- Text must be single-spaced spaced Arial 11 with 2cm margins. You cannot include more text by using smaller fonts or narrower margins.
- You must present your ideas for all specified sections
- References must be included within the page limits.
- References and citations must be in a consistent format. The Harvard referencing style is recommended.
- You must submit via Minerva one copy of the report as a PDF document.

© University of Leeds

The quality of written English will be assessed in this work. As a minimum, you must ensure:

- Paragraphs are used
- There are links between and within paragraphs although these may be ineffective at times
- There are (at least) attempts at referencing
- Word choice and grammar do not seriously undermine the meaning and comprehensibility of the argument
- Word choice and grammar are generally appropriate to an academic text

These are pass/ fail criteria. While marks are not awarded to these, if do not meet these criteria you will fail overall.

In addition, your submission should adhere to the guidelines on the presentation of coursework as set out in **Section 3.2a of the Code of Practice on Assessment**.

Word count

There is a strict **page limit of 12 pages** for this assessment in total. See the 'Task' section above for more information on this.

The word limit is an extremely important aspect of good academic practice and must be adhered to. Your submission should adhere to the guidelines set out in Section 3.3d of the Code of Practice on Assessment. You are required to state an accurate word count on the cover page of your assignment. If the amount of work submitted is higher than that specified by the word limit or that declared on your word count, this may be reflected in the mark awarded and noted through individual feedback given to you.

Referencing

It is important that you show that you have engaged with appropriate literature and make efforts to explicitly cite all the sources that you have used or have shaped your thinking.

All sources that you draw on in your submission should be correctly referenced using the Harvard style. Please see the Library webpages for guidance.

Academic Integrity

© University of Leeds

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Submit your Report by the due date and time via Turnitin using the submission link in Minerva.

Your student ID number should be included in the document, in the file name, and entered as the 'submission title' when you upload your work via Turnitin.
Please include a cover sheet when you submit your work.

Marking

This is a compulsory assessment for OCOM5204M Data Mining and Text Analytics. It is worth **70%** of your overall module grade.
The mark scheme for your report will reflect the EPSRC grading scheme with marks in the range 0-6 for each criterion, EXCEPT the sections on Pilot study, Programme and methodology, and Appendix. These are most important section in the report, and therefore have more marks available for them:
Background (0-6 marks)
Importance and contribution to knowledge (0-6 marks)
Research hypothesis and objectives (0-6 marks)
Pilot study (0-12 marks)
Programme and methodology, with corresponding workplan diagram (0-12 marks)
Appendix: use of data mining and text analytics tools in developing your Report (0-18 marks)
TOTAL: up to 60 marks

Feedback

© University of Leeds

You will receive your mark and feedback via Minerva, in the **Gradebook** section, accessed from the tabs at the top of the page. We will notify you by email when your feedback is available. You will usually receive feedback within 15 working days of the assignment deadline. This feedback may take the form of a written or recorded audio comment.

© University of Leeds

Reading List

The reading list and online course readings for this module can be found in the 'Reading List' section of Minerva.

Additional Information and Support

Key Documents

Please ensure that you read and familiarise yourself with the following important documents, available in the Sustainable Business Leadership Online organisation area on Minerva:

- Digital Education Service (DES) Code of Practice on Assessment
- Code of Practice on Attendance

Skills@Library

Skills@Library provides support for students to develop academic skills. This includes areassuch as researching, writing, note making and referencing. This support is delivered throughonline resources, workshops and webinars: http://library.leeds.ac.uk/skills.

General Module Forum

The Digital Education Service and the University of Leeds are committed to supporting students. If you have any questions about the academic content of the module, please post them in the module's 'General Module Forum' discussion board. You can access this throughthe 'Discussion Board' section of Minerva. The Module Team will reply to your post and yourpeers will also have access to the answer.

Student Success Advisor

If you have any questions about using Minerva, administrative processes, or pastoral care please contact your Student Success Advisor. They will be able to advise you or signpost youto the correct resource.

© University of Leeds

Recording Policy

As part of your study on this module, webinars will be recorded by the University. This recording is conducted in accordance with the University's Policy on Audio or Video Recordingfor Educational Purposes. Please read the Policy for further background information relating toyour rights. The purpose of the recording is to support your study at the University. It will be accessible by students enrolled on the module through Minerva.

In accordance with the University's Intellectual Property Policy in general where staff or students make a recording, all rights in that recording (including rights in the sounds made) willbelong to the University. The University seeks to respect student and staff rights to be acknowledged as authors and performers. Otherwise, to the extent allowed under the general law, any individual interests in the recordings are waived, allowing the University to act in effectas a custodian, to maximise educational objectives in the general interest of all.

If you have any concerns relating to the recording please contact the individual leading the educational activity, or the module leader. If you wish to opt out of being recorded or wish to request that your contribution be edited please contact the individual leading the activity or themodule leader in advance of the session (for optouts) or immediately after the session (for requests for edits). Please note that you may not opt out of recordings which form part of the assessment process on the module. If you have any concerns relation to the recording, please contact your Student Success Advisor.

Complaints

If you have any complaints about this module, please write to DESStudents@leeds.ac.uk andthe Digital Education Service will review your complaint. If a remedy cannot be arrived If a remedy cannot be arrived at, the University's official complaints procedure could be followed.

© University of Leeds